

Cycle 1 Clone 0 Report — RC7 v2 Rerun on c51 Branch A+B Substantive MIDIs (Fork 18817b483ed4)

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[OUTPUT: report_cycles_1-1_clone_0]

Cycle 1 Clone 0 Report — RC7 v2 Rerun on c51 Branch A+B Substantive MIDIs (Fork 18817b483ed4)

Abstract

Cycle 1 of clone-0 (fork 18817b483ed4) lands the c53 Branch A substantive RC7 v2 rerun for M-RECREATE-2/accurate-small-set/rc7-mix-balance-match at **RC7_v2_LANDS**. The c51 Branch C RC7 mix-balance pipeline was re-run against c51 Branch A+B substantive per-stem MIDIs (vocals+guitar+piano from data/rc1_rc9_impl/*/merged_partial.midi; drums+bass from data/rc2_rc3_impl/*/merged.midi) replacing the c33-anchor placeholder MIDIs used at c51. **5/5 focus songs pass A7, 20/20 stem accepts across {drums, bass, other_guitar, other_piano}**. Chicken Grease (31a164f845f8e27e, band 6) passes 4/4 stems — the c52 handoff outcome confirms the c51 Branch C mechanism was sound: c51's RC7_FAILS was purely upstream MIDI under-transcription, not an EQ/loudness-match pipeline defect. Byte-determinism × 2 across all 226 output files under pinned environment. Auditor decision: **COMPLETE** with [[BRANCH_

COMPLETE]].

Verdict

RC7_v2_LANDS (VALIDATED; 5/5 focus songs pass A7 across all four stems; 20/20 individual stem accepts — well above the RC7_v2_LANDS threshold of ≥ 3 songs).

Rubric SHA Anchor Chain (Three-Way Byte-Equal + Parent-v2 Preserved)

Location	SHA-256
docs/rc7_v2_rerun_rubric.md	9f24e6d9...04dde4
data/recreate_v2/rc7_out_v2/rubric_hash.txt	9f24e6d9...04dde4
verdict.json.rubric_hash	9f24e6d9...04dde4
c50 v2 parent rubric (docs/m_recreate_2_accurate_small_set_rubric_v2.md)	0e11f704...debe1f (independent chain, byte-preserved)
c49 v1 rubric	958ade38...3fe58b9d (byte-preserved)

Three-way byte-equality chain CONFIRMED. Fresh per-milestone rubric hash resolves cleanly independent of c50 v2 doc SHA.

Pre-Registration mtime Discipline (Clean Ordering)

- Rubric mtime 1788035291 < impl mtime 1788035649.
- Rubric pre-registered before any Python edit under `scripts/recreate_v2/rc7_v2_*.py` (mtime-hard gate honored per c46 path (ii) amendment).

READ-ONLY Anchor Preservation (c51 Branch C + render_stem)

Anchor	SHA-256 / State
scripts/palette_render/render_stem.py	214372d920a319a97d6e3fc7b9ee4134c08c0cb4aecb776f4a50c75f965b5 byte-identical pre/post (directive-anchored SHA)
data/recreate_v2/rc7_out/ (c51 Branch C)	182 files byte-identical pre==post per anchor-preservation-v2 ledger event
data/rc1_rc9_impl/ (c51 Branch A)	READ-ONLY (worker did not touch c51 anchors)
data/rc2_rc3_impl/ (c51 Branch B)	READ-ONLY (worker did not touch c51 anchors)

Consumed `render_stem.py` in its c51-extended additive-kwargs form (`eq_curve + loudness_target`) READ-ONLY.

RC7 v2 Per-Song Results (5/5 Focus Songs Pass A7; 20/20 Stem Accepts)

Per focus song: fit 12-band iirpeak EQ curve ($Q=1.4$, log-spaced) from original 6-stem spectrum vs new bare-render (built from substantive MIDIs); applied RMS + LUFS-S

loudness match to reach A7 (≤ 3 dB RMS, ≤ 3 LU LUFS-S vs baseline/<sha16>/rc7_per_stem_loudness.json); summed matched stems into data/recreate_v2/rc7_out_v2/<sha16>/rc7_v2_mixed_reconstruction.wav.

Chicken Grease (31a164f845f8e27e, band 6, mandatory per operator override): 4/4 stems pass — kill-shot on c51's placeholder-MIDI shortfall. Substantive Branch A+B MIDIs deliver the EQ+loudness match cleanly.

5 songs × 4 stems = 20/20 individual stem accepts.

Byte-Determinism × 2 (226/226 SHA-Equal)

226/226 files SHA-256 equal across two fresh tempfile.mkdtemp() runs. Env pins present: - PYTHONHASHSEED=0 - SOURCE_DATE_EPOCH=1756463424 - TZ=UTC - LC_ALL=C.UTF-8 - Single-thread BLAS

D4 Old c33 Chorus+Reverb Chain (Preserved as Diagnostic ONLY)

d4_old_chain_baseline_present=true; per-song panel_baseline_old_chain_v2.tsv preserved as diagnostic ONLY. Never a LANDS deliverable (per c50 v2 discipline invariant).

Test Surface (All Green)

Suite	Result
promise_check	0 ERRORS , WARNs all pre-existing (unchanged from c52 baseline)

- test_04: NO PRNG grep-guarded (PASS).
- test_08: c48 env-var flags default OFF (PASS).

Ledger Events (9; Lines 880-888; Correct c32/c33 Suffix Discipline)

Landed at ledger lines 880-888 (tail = 888 rows):

- **Substantive M-RECREATE-2/accurate-small-set/rc7-mix-balance-match/* unsuffixed per c32** (6 named events):
 - rubric committed, pre-registration verified, impl landed, byte-determinism-v2 verified, anchor-preservation-v2 verified, verdict rollup
- **Infra + housekeeping -clone-0 suffixed per c33** (2 events):
 - _infra/adopt-rc7-v2-artifacts-clone-0, _archive/rc7-v2-scratchpad-clone-0
- **M-INGEST-1/egress-probe-cycle53-clone-0** (1 event at tail; 429 + tv_embedded unchanged)

Ledger-event budget: 6 named + 2 housekeeping + 1 egress = 9 events, under the directive's stated budget.

MODERATE Findings (0 Blocking; All Discipline Invariants Met)

- data/rc1_rc9_impl/ + data/rc2_rc3_impl/ READ-ONLY (worker did not touch c51 anchors).
- c48 env-var flags default OFF (test 08).
- NO PRNG (test 04 grep-guarded).
- /usr/bin/python3 guard present.
- VST3 lock (c35 anti-pattern) and CLAP anti-pattern (c11, VGGish DEFERRED-None) respected.
- Substantive sub-leaves unsuffixed per c32; infra -clone-0 suffixed per c33.

MINOR Findings (Logged, Not Acted On)

- ~20 pre-existing `promise_check` WARNs unchanged from c52 (drift-tracking, out of scope per c48 auditor precedent).
- Session housekeeping registered `_plan/register-c53-rc7-v2-fanout-milestones + 6 plan_of_record` rows retroactively — appropriate given the deliverable was already on disk from a prior c53 pass.
- **Session-scope disclosure:** `research_brief` fed to this session was for c53-clone-1 (RC1 policy pivot), while directive was for c53-clone-0 (RC7 v2 rerun). Worker correctly followed the directive. Harness-wiring artefact worth noting for campaign meta-audit; does not affect this clone's deliverable.

Merge Report Path Fallback (Empirically Confirmed Necessary)

Merge report written to **in-project fallback path** `reports/fanout/fork-18817b483ed4/clone-0/merge_report.md` (4,666 B) because the directive-named path `/home/user/music-gen-instance/...` is outside session Directory Boundaries. Worker correctly applied the brief's Priority 4b Option (i) recommendation.

Root conductor should poll the in-project path.

The outside-boundaries merge path deadlocks without the in-project fallback. Priority 4b (`_infra/fanout-merge-report-path-in-project`) is empirically confirmed necessary this cycle — should escalate to CRITICAL when re-issued.

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	RC7 v2 rerun against c51 Branch A+B substantive MIDIs; 5-song focus set; A7 gate	Rubric pre-registered (mtime-hard); 12-band iirpeak EQ + RMS + LUFS-S loudness match; per-song reconstruction WAV; byte-det × 2 across 226 files; 9 ledger events; merge report at in-project fallback	COMPLETE with <code>[[BRANCH_COMPLETE]]</code>

State-Machine Discipline (c29 Lemma Respected)

- `M-RECREATE-2/accurate-small-set/rc7-mix-balance-match/*` is a peer sub-leaf under c50 v2 rubric chain. NOT a child of any terminal-validated ancestor.
- Peer-supersede pattern from c50 preserved: c49 v1 rubric + c50 v2 rubric + c53 clone-0 rubric all byte-preserved on their own chains.
- No validated → `in_progress` transitions attempted.

Sub-Topic Assessment (16/16 Directive Acceptance Criteria Met)

Criterion	Status
Rubric pre-registered mtime-hard	☐
render_stem.py SHA byte-identical pre/post	☐
12-band iirpeak EQ (Q=1.4, log-spaced)	☐
RMS + LUFS-S loudness match	☐
Sum to rc7_v2_mixed_reconstruction.wav per song	☐ (5 present)
Byte-det × 2 with env pins	☐ (226/226)
D4 old chain preserved as diagnostic ONLY	☐
Three-way rubric_hash-v2 chain	☐
c51 rc7_out anchor 95+ SHAs byte-identical	☐ (182 files)
VST3 lock (c35) + CLAP (c11) respected	☐
NO PRNG	☐
/usr/bin/python3 guard	☐
c51 anchors READ-ONLY	☐
c48 env-var flags default OFF	☐
Egress probe M-INGEST-1/egress-probe-cycle53-clone-0 at tail	☐
Required output docs/rc7_v2_rerun_report.md	☐ (7,625 B)

All 16 criteria met.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908 (not relevant to this branch).
- SHA-256 tiebreak; no PRNG; no sidecar_nonfactor; no i4_stratified; no render_effects_layered (retired).
- Interpreter guard `#!/usr/bin/python3` on every new script.
- Read-only anchors preserved: c14 _ledger_schema.py; c22 stability harness; c26 Path B commitment; c31/c33/c34/c35/c36/c37/c45/c46/c47/c50/c51 palette + recreate + anchor-manifest + rubric chain; render_stem.py (Branch C anchor) byte-identical.
- Rated audio egress-blocked at *.googlevideo.com (429 + tv_embedded unchanged; M-INGEST-1/egress-probe-cycle53-clone-0 recorded honestly).
- Ledger hygiene: narrative field; run_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses UUID5 content-hash event_id; two-arg append_ledger_event(workspace, event).
- **c48 env-var flags default OFF** (MUSICGEN_LEDGER_SUBSTANTIVE_EXEMPTION, MUSICGEN_LEDGER_SUPERSEDES_IN_HASH).

Anti-Patterns Locked (5-Count Stable)

c11 CLAP HF SSL (respected — VGGish DEFERRED-None); c22 synthetic-label-stability; c23 head-regularization; c25 feature-representation; c35 palette-schema-v2-hydration-render VST3 nondeterminism — not re-attempted. c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged. c31 STILL_GAP surface intact.

No M-EAR-1/* or M-GEN-1/* emissions this branch.

Merge Disposition

Merge report at reports/fanout/fork-18817b483ed4/clone-0/merge_report.md (4,666 B) per in-project fallback. Root conductor should poll the in-project path.

Ledger tail = **888 rows**; promise_check 0-ERROR; 9 c53 clone-0 events landed at lines 880-888 with correct c32/c33 suffix discipline.

Cycle-54+ Handoff (Per Cycle-1 Auditor Guidance)

1. **RC7 v2 mechanism is proven on 5 focus songs**; extending to a broader corpus is a candidate next scope but must NOT block on egress (unchanged HTTP 429 + tv_ embedded).
2. **Formalise _infra/fanout-merge-report-path-in-project** as an infra milestone in c54 — the outside-boundaries merge path deadlocks without the in-project fallback; Priority 4b (from c53 clone-1 brief) empirically confirmed CRITICAL this cycle.
3. **Auditor-reads-ledger-not-brief-summaries lemma** track record continues: caught worker-report drift at c48-close AND c49-close; independent verification at c50 + c51 + c53. Codification in docs/auditor_discipline_ledger_first.md remains recommended for c54.
4. **c53 fanout merge conductor tasks**: reconcile clone-0 (RC7 v2, this branch) with sibling clones (clone-1 RC1 policy pivot; clone-2 if any) shadow ledgers under _infra/fanout-concat-hardening invariants.
5. **c48 env-var flag flips** remain c54+ candidates post-M-RECREATE-2 arc closure.
6. **Session-scope harness-wiring artefact** (research_brief was for c53-clone-1 while directive was for c53-clone-0): worth noting in campaign meta-audit; does not affect this clone's deliverable.

Standing follow-ons (carried from prior cycles):

7. **c48 Branch A _infra/harness-and-writer-hardening-v3** still owed as substantive re-field.
8. **Writer-side guard rejecting novel milestone_ids not in plan_of_record** — plan-registration-lag pattern.
9. **_infra/large-model-fetchability-registry** — c50's htdemucs_6s OK; if VGGish also fetches cleanly, argues for pinning.
10. **Mura Masa RC3 threshold-boundary refinement** (from c51 clone-1) — c52+ candidate.
11. **RC5 full implementation with octave normalisation** — c52+ candidate.
12. **RC6-v2 panel gate** — c52+ candidate (depends on VGGish availability).

Cumulative Progress

M-RECREATE-2 arc (post-c53 clone-0 substantive advance):

Cycle	Milestone	Verdict / Status
c49	M-RECREATE-2/accurate-small-set (v1)	rubric committed; substantive close
c50	M-RECREATE-2/accurate-small-set-v2 (peer supersede)	rubric committed; htdemucs_6s fetch OK; 5-song chosen_section (fork sibling)
c51 clone-0 (A)	M-RECREATE-2/.../{rc1-vocals,rc9-first-class-parts}-transcription	
c51 clone-1 (B)	M-RECREATE-2/.../{rc2-drum-onset,rc3-bass}-transcription	RC2_RC3_LANDS (5/5 RC2; 4/5 RC3; Chicken Grease kill-shot)
c51 clone-2 (C)	M-RECREATE-2/.../rc7-mix-balance (c33-anchor placeholder MIDIs)	RC7_FAILS (upstream MIDI under-transcription)
c52	LINEAR integration	(integration + RC5 full + RC6-v2 readiness)

Cycle	Milestone	Verdict / Status
c53 clone-0 (A) (this)	M-RECREATE-2/.../rc7-mix-balance-match (substantive Branch A+B MIDIs)	RC7_v2_LANDS (5/5 songs pass A7; 20/20 stem accepts; Chicken Grease 4/4 stems)

The c51 RC7_FAILS verdict was purely upstream MIDI under-transcription, not a defect in the EQ/loudness-match pipeline — c53 clone-0 empirically confirms the c51 Branch C mechanism was sound. Substantive Branch A+B MIDIs (from c51 clone-0/clone-1 substantive advances) deliver the EQ + loudness match cleanly.

Chicken Grease result: 4/4 stems passing on substantive Branch A+B MIDIs after failing on c51 c33-anchor placeholders — expected c52 handoff outcome; validates the operator's insistence that Chicken Grease band-6 be treated as mandatory anchor.

Fanout cadence: LINEAR c49/c50 → FANOUT c51 (A/B/C) → LINEAR c52 (integration) → **FANOUT c53 (this branch is Branch A: RC7 v2 rerun)** → LINEAR c54 recommended (fanout merge integration; broader corpus consideration).

Anti-pattern discipline: 5 confirmed campaign anti-patterns (c11 CLAP, c22 chassis-audit, c23 head-reg, c25 feature-rep, c35 palette-v2 VST3) all remain respected across c53 clone-0.

Rubric mtime discipline: c46 path (ii) amendment (mtime-hard, git-log advisory) continues to work correctly under fanout — rubric mtime 1788035291 cleanly pre-dates impl mtime 1788035649.

Auditor-reads-ledger-not-brief-summaries lemma (proposed c50) confirmed relevant here: worker's report correctly disclosed the brief/directive mismatch AND the merge-path deadlock, both of which auditor verified independently on-disk. Track record: caught worker-report drift at c48-close AND c49-close, and independent verification at c50 + c51 + c53.

Merge-report path deadlock: empirically confirmed this cycle. Real infra defect that will silently break any fanout where conductor is not aware of in-project fallback. Priority 4b escalates to CRITICAL when re-issued in c54.

c29 state-machine lemma respected: peer sub-leaves under c50 v2 rubric chain; ledger topology stays a DAG.

c32 → c33 → c36 v2 → c39 v3 → c47 Branch B MIXED → c50 peer-supersede fanout-namespace + rubric-chain convention held: substantive M-RECREATE-2/* unsuffixed; infra families -clone-0 suffixed.

Anchor-manifest arc: v1.1 stable at 19 entries with SOURCE_DATE_EPOCH first-class.

Collision-modeling arc: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Egress state: 429 + tv_embedded unchanged for playlist audio; htdemucs_6s model-weight fetchability OK (c50 positive) distinct from playlist harvest.

Scope of this fanout clone fully discharged. [[BRANCH_COMPLETE]] emitted; auditor decision **COMPLETE**.

[END OUTPUT]